Dataset Description — Chest X-Ray Pneumonia

1 Overview

The **Chest X-Ray (Pneumonia)** dataset is a large-scale medical imaging dataset designed for binary classification: Pneumonia Detection from Chest X-Ray Images. It contains anterior-posterior chest X-ray images of patients diagnosed with pneumonia and healthy individuals. The dataset is widely used in computer vision and deep learning research for medical diagnostics.

2 Dataset Source

- **Origin:** Kaggle Dataset Repository
- **URL:** https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia
- **Author:** Paul Mooney
- **Institution:** Guangzhou Women and Children's Medical Center, China

All chest X-ray images were screened by expert radiologists to confirm the diagnosis.

3 Dataset Structure

After preprocessing, the dataset is organized as follows:

```
data/├──
 train/│├──
    NORMAL/│└──
    PNEUMONIA/├──
 val/│├──
    NORMAL/│└──
    PNEUMONIA/└──
 test/├──
     NORMAL/└──
     PNEUMONIA/
```

Each folder contains grayscale chest X-ray images in `.jpeg` or `.png` format.

4 Class Distribution

Split	NORMAL	PNEUMONIA	Total
Train	∼1,083	∼2,778	∼3,861
Val	∼135	∼336	∼471
Test	∼234	∼390	∼624
Total	**1,452**	**3,504**	**4,956**

Table 1: Class distribution across dataset splits.

Pneumonia cases outnumber normal cases by roughly **2.4**×, which introduces **class imbalance** handled via weighted sampling in training.

5 Image Properties

- **Format:** JPEG / PNG
- **Color Space:** Grayscale (1 channel)
- **Average Resolution:** 1024×1024 pixels (varies slightly)
- **Resized to:** 224×224 pixels before training
- **Normalization:** Pixel values scaled to [0, 1] range

6 Preprocessing Steps

The preprocessing pipeline includes:

1. **Download** from Kaggle using the API
2. **Extraction** of the original “train”, “val”, “test” folders
3. **Merge and Re-split** with an 80/10/10 ratio
4. **Resize** images to (224×224)
5. **Normalization** and conversion to PyTorch tensors

All steps are automated via the script `src/download_data.py`.

7 Intended Use

This dataset is used to train deep learning models that can:

- Detect pneumonia from X-ray scans

- Assist radiologists in early diagnosis
- Evaluate deep learning architectures (CNNs, ResNets, Transformers, etc.)

The dataset is suitable for **binary image classification** and **medical image analysis** research.

8 Citation & License

Citation:

> Kermany, Daniel, et al. "Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning." *Cell*, 2018.

License: The dataset is provided under Kaggle's data terms for educational and research purposes.

9 Summary

Property	**Description**
Task	Pneumonia detection from chest X-ray
Classes	NORMAL, PNEUMONIA
Images	4,956 total
Image Type	Grayscale, frontal chest X-ray
Use Case	Binary classification, medical imaging
Recommended Model	CNN or ResNet18
Preprocessing	Resize (224×224), Normalize, Tensor conversion

Table 2: Summary of dataset properties.